

$$\dot{\vec{P}} = \vec{F}_{ext}, \text{ if } \vec{F}_{ext} = 0$$

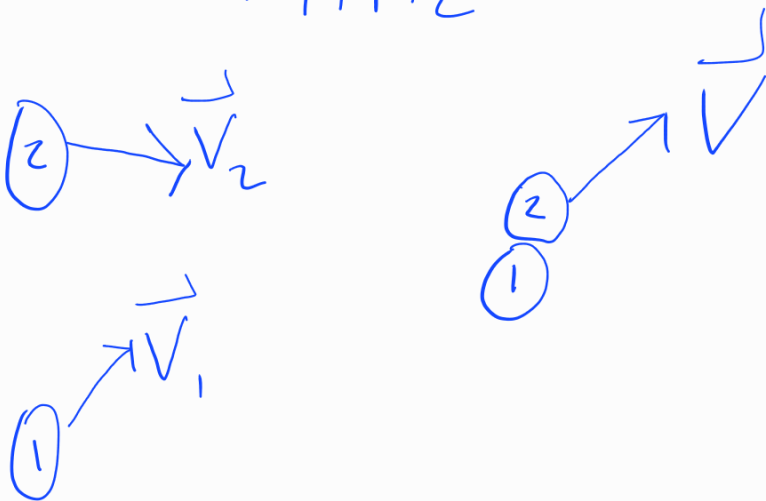
$$\vec{P} = \sum m_{\alpha} \vec{V}_{\alpha}$$

Ex 3.1 (2 Body Perfectly inelastic)

$$P_i = m_1 \vec{V}_1 + m_2 \vec{V}_2$$

$$P_f = m_1 \vec{V} + m_2 \vec{V} = (m_1 + m_2) \vec{V}$$

$$\vec{V} = \frac{m_1 \vec{V}_1 + m_2 \vec{V}_2}{m_1 + m_2}$$



If  $\vec{V}_2 = 0$ ,

$$\vec{V} = \frac{m_1}{m_1 + m_2} \vec{V}_1$$

Rockets

$$P(t) = mV$$



$$\begin{aligned} P(t+dt) &= (m+dm)(v+dv) - dm(v-V_{ex}) \\ &= mV + m dv + dm v_{ex}, \quad dm dv \approx 0 \end{aligned}$$

$$dP = P(t+dt) - P(t) = m dv + dm v_{ex}$$

$$dP = 0, \quad F_{ext} = 0$$

$$m dv = -dm v_{ex}, \quad \text{divide by } dt$$

$$m \dot{v} = -\dot{m} v_{ex}$$

Thrust:

$$\text{thrust} = -\dot{m} v_{ex} \quad , \quad (\dot{m} \text{ is negative})$$

$$m dv = -dm v_{ex}$$

$$dv = -v_{ex} \frac{dm}{m}$$

$$\int dv = -v_{ex} \int \frac{dm}{m}$$

$$V - V_0 = v_{ex} \ln\left(\frac{m_0}{m}\right)$$

# Center of Mass



$$\vec{R} = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} \vec{r}_{\alpha} = \frac{m_1 \vec{r}_1 + \dots + m_N \vec{r}_N}{M}$$

$$M = \sum m_{\alpha}$$

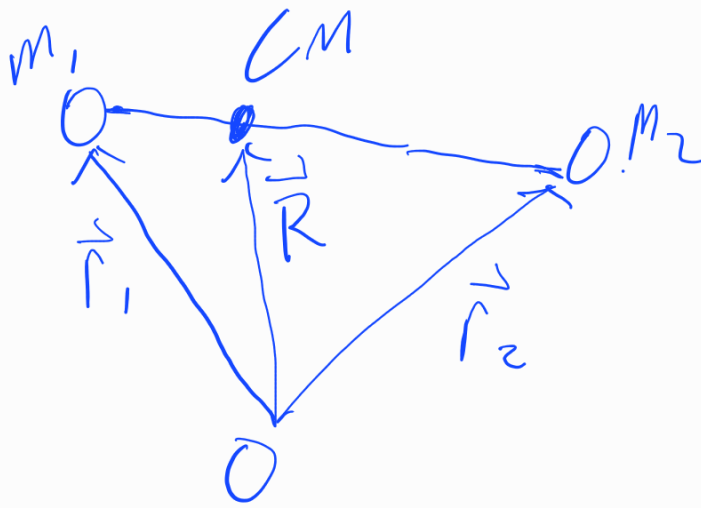
$$X = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} x_{\alpha}, \quad Y = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} y_{\alpha},$$

$$Z = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} z_{\alpha}$$

$$N=2$$

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

if  $m_1 \approx M$ ,  $m_\alpha \ll M$



Momentum in terms of CM

$$\vec{P} = \sum_{\alpha} \vec{p}_{\alpha} = \overbrace{\sum_{\alpha} m_{\alpha}}^M \dot{\vec{r}}_{\alpha} = M \dot{\vec{R}}$$

Extending to:

$$F_{\text{ext}} = \dot{P} = M \ddot{R}$$

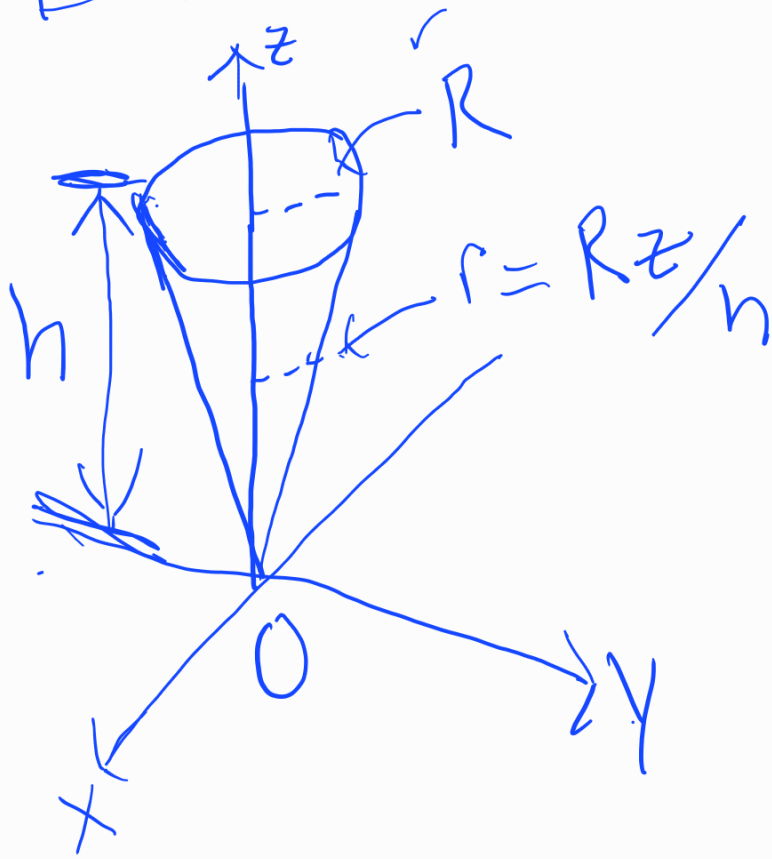
CM for continuous mass distribution:

$$\vec{R} = \frac{1}{M} \int \vec{r} dm = \frac{1}{M} \int \underbrace{\rho}_{\text{density}} \underbrace{\vec{r} dV}_{dm}$$

EX 3.2:

Solid cone

symmetric  
about  $z$



$$\therefore \vec{R} = \vec{z}_{cm}$$

$$z_{cm} = \frac{1}{M} \int \rho z \, dV = \frac{\rho}{M} \int z \, dx \, dy \, dz$$

$x, y$  bound to circle w/  $r = Rz/h$  so

$$\pi r^2 = \pi \frac{R^2 z^2}{h^2} = dx \, dy$$

$$\text{So, } z_{cm} = \frac{\rho \pi R^2}{M h^2} \int_0^h h^3 \, dz$$

$$Z_{cm} = \frac{\rho \pi R^2}{M h^2} \frac{h^4}{4}$$



$$M = \rho V_{\text{cone}} = \frac{1}{3} \rho \pi R^2 h$$

$$\therefore Z_{cm} = \frac{3}{4} h$$

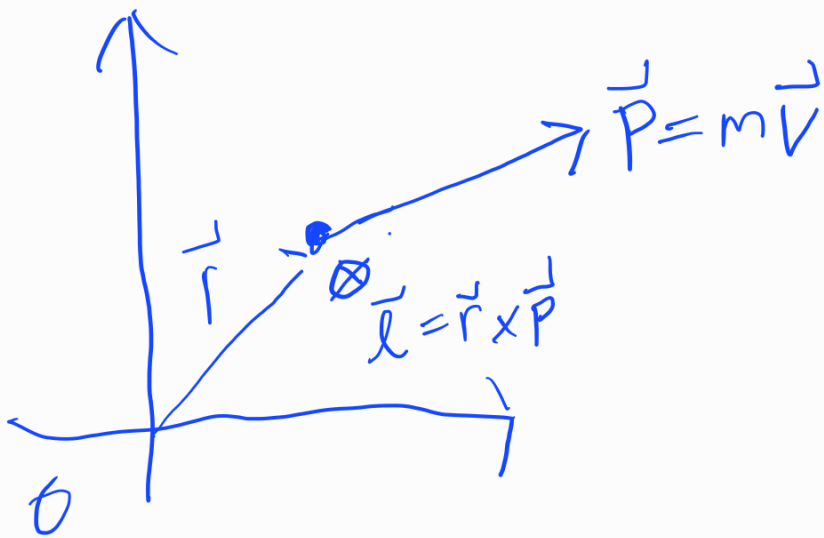
# Angular Momentum (single particle)

$$\vec{l} = \vec{r} \times \vec{p}$$

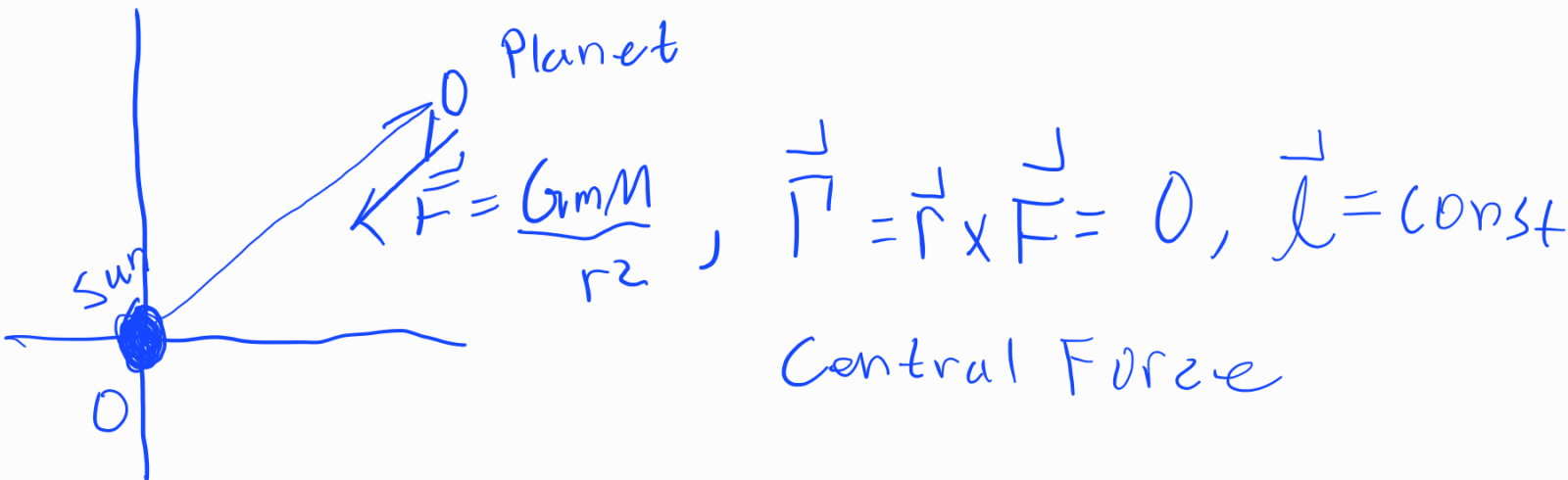
$$\dot{\vec{l}} = \frac{d}{dt}(\vec{r} \times \vec{p}) = (\dot{\vec{r}} \times \vec{p}) + (\vec{r} \times \dot{\vec{p}})$$

$$= (\cancel{\dot{\vec{r}} \times m\dot{\vec{r}}}) + (\vec{r} \times \dot{\vec{p}})$$

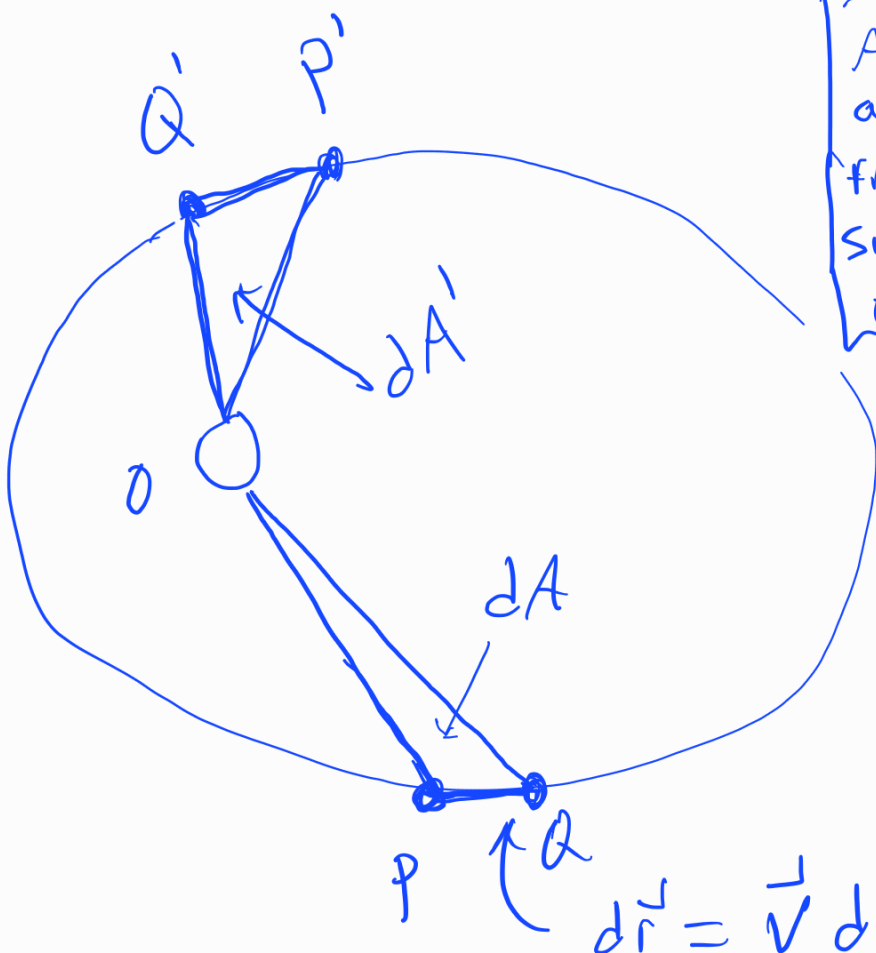
$$\dot{\vec{l}} = \vec{r} \times \vec{F}_{\text{ext}} \equiv \vec{\tau}$$



# Kepler's Second Law:



Since  $\vec{r} \times \vec{p}$  is constant ( $\vec{L} = \text{const}$ )  
 $\vec{r}$  &  $\vec{p}$  must be constrained to a  
 single plane



As each Planet moves  
 around the sun, a line drawn  
 from the Planet to the sun  
 sweeps out equal areas in  
 equal times \*

$$dt = dt'$$

$$d\vec{r} = \vec{v} dt$$

$$dA = \frac{1}{2} |\vec{r} \times \vec{v} dt|, \text{ Area of triangle OPQ}$$

replace  $\vec{v}$  by  $\frac{\vec{p}}{m}$  & divide by  $dt$ ,

$$\frac{dA}{dt} = \frac{1}{2m} |\vec{r} \times \vec{p}| \frac{dt}{dt} = \frac{1}{2m} \vec{l}$$

$\vec{l}$  is a const.

$$\therefore \frac{dA}{dt} \text{ is const}$$

# Angular Momentum for Several Particles:

$N$  particles,  $\alpha = 1, 2, 3, \dots, N$

$$\vec{l}_\alpha = \vec{r}_\alpha \times \vec{p}_\alpha \quad (\text{all } \vec{r}_\alpha \text{ measured from } O)$$

Total Angular Momentum  $\vec{L}$

$$\vec{L} = \sum_{\alpha=1}^N \vec{l}_\alpha = \sum_{\alpha=1}^N \vec{r}_\alpha \times \vec{p}_\alpha$$

$$\dot{\vec{L}} = \sum_{\alpha} \dot{\vec{l}}_\alpha = \sum_{\alpha} \vec{r}_\alpha \times \vec{F}_\alpha$$

$$\vec{F}_\alpha = \sum_{\beta \neq \alpha} \vec{F}_{\alpha\beta} + \vec{F}_\alpha^{\text{ext}}$$

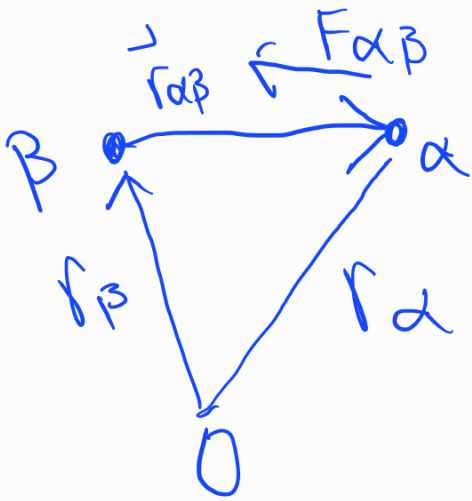
$$\dot{\vec{L}} = \sum_{\alpha} \sum_{\beta \neq \alpha} \vec{r}_\alpha \times \vec{F}_{\alpha\beta} + \sum_{\alpha} \vec{r}_\alpha \times \vec{F}_\alpha^{\text{ext}}$$

Remember  $F_{21} = -F_{12}$

$$\sum_{\alpha} \sum_{\beta \neq \alpha} \vec{r}_{\alpha} \times \vec{F}_{\alpha\beta} = \sum_{\alpha} \sum_{\beta \neq \alpha} (\vec{r}_{\alpha} \times \vec{F}_{\alpha\beta} + \vec{r}_{\beta} \times \vec{F}_{\beta\alpha})$$

$$= \sum_{\alpha} \sum_{\beta \neq \alpha} (\vec{r}_{\alpha} - \vec{r}_{\beta}) \times \vec{F}_{\alpha\beta}$$

Look at  $(\vec{r}_{\alpha} - \vec{r}_{\beta}) = \vec{r}_{\alpha\beta}$



$\therefore$  all  $F_{\alpha\beta}$  are central forces

$$\vec{r}_{\alpha\beta} \times \vec{F}_{\alpha\beta} = 0$$

$$\dot{L} = \vec{\Gamma}_{ext}$$

If  $\vec{\Gamma}_{\text{ext}} = 0$  then

$\vec{L} = \sum \vec{r}_\alpha \times \vec{p}_\alpha$  is constant

Conservation of angular momentum

Moment of Inertia:

from general Phys

$$L = I\omega, \quad I = \text{moment of inertia}$$

Disk (center rotation)

$$I = \frac{1}{2}MR^2$$

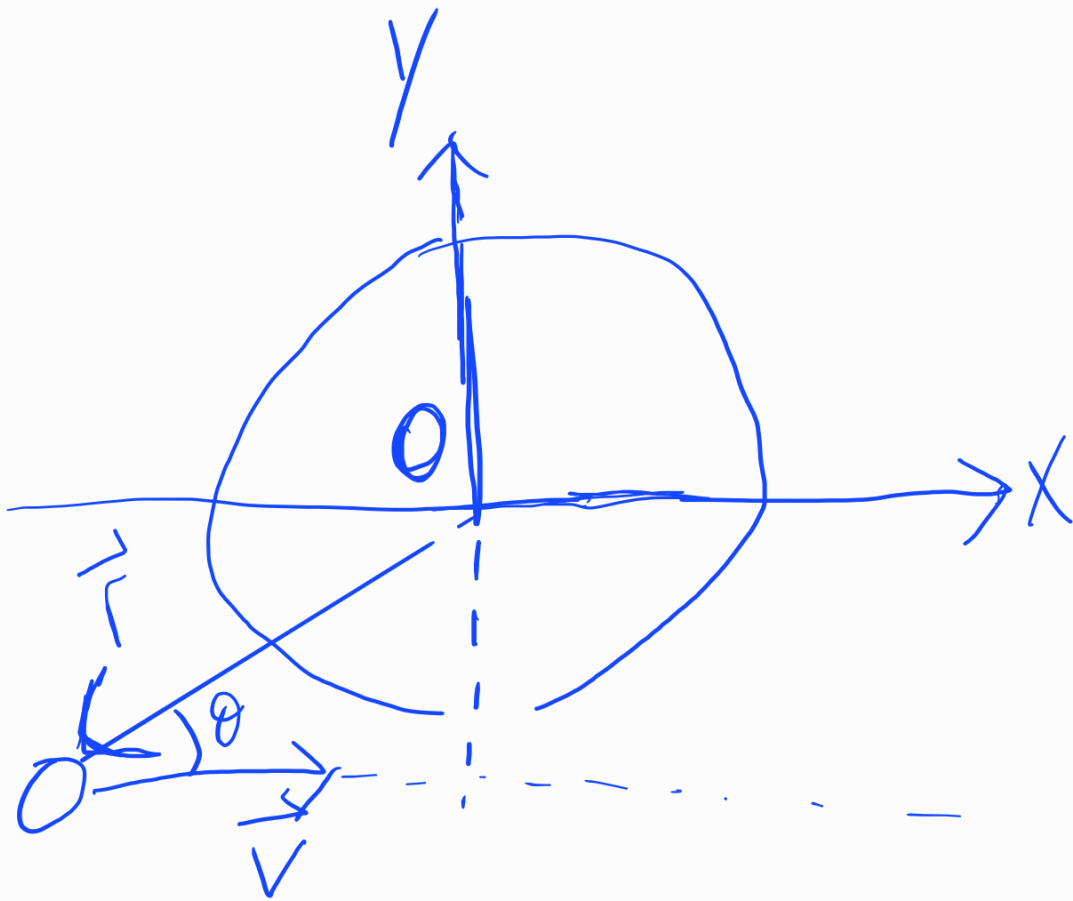
Sphere

$$I = \frac{2}{5}MR^2$$

In general (mult-Particle)

$$I = \sum_1 m_{\alpha} \rho_{\alpha}^2, \quad \rho_{\alpha} = \text{distance from axis of rotation}$$

Ex 3.3 Collision of a Lump of Putty w/ Turntable



$$\vec{L} = 0 \quad L_z = \text{Const.}$$

$$L_{\text{putty}} = \vec{r} \times \vec{P}[\hat{z}]$$

$$L_{z_0} = L_z = r(mv) \sin \theta \\ = mvr_b$$

$$I = (m + \frac{M}{2}) R^2$$

$$L_{z_f} = I \omega$$

$$L_{z_0} = L_{z_f}$$

$$m v b = \left(m + \frac{M}{2}\right) R^2 \omega$$

$$\omega = \frac{m}{\left(m + \frac{M}{2}\right)} \cdot \frac{v b}{R^2}$$

Angular Momentum about the  
CM

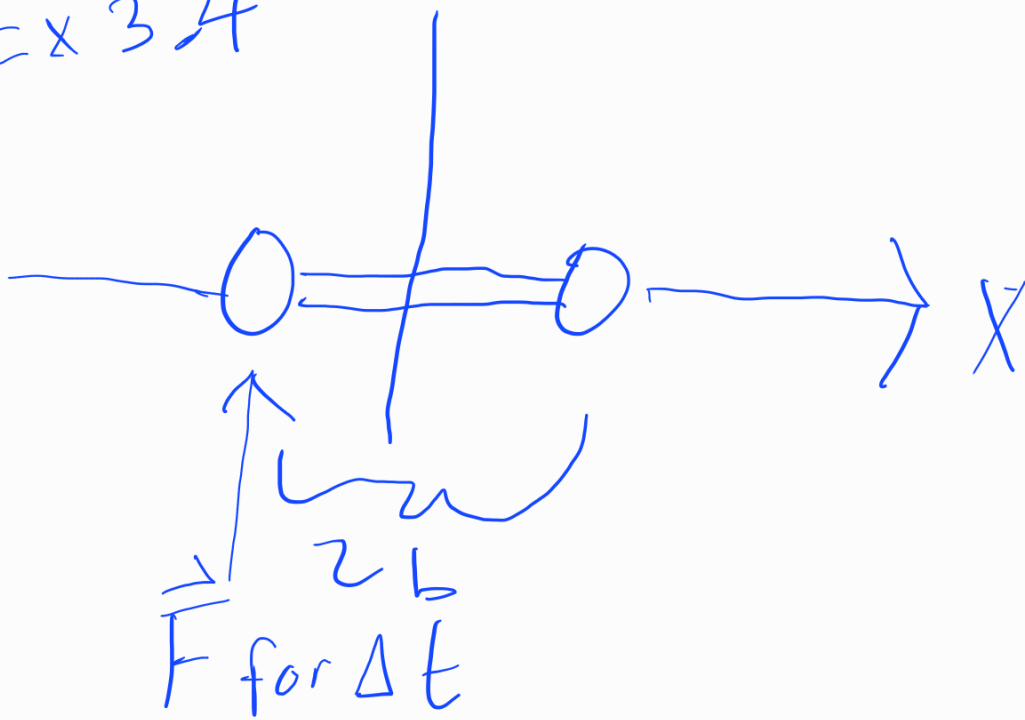
$$\dot{\mathbf{L}} = \mathbf{\Gamma}_{\text{ext}}$$

$$\frac{d}{dt} \mathbf{L}(\text{about CM}) = \mathbf{\Gamma}_{\text{ext}}(\text{about CM})$$

if  $\mathbf{\Gamma}_{\text{ext}}^{\text{CM}} = 0$  ,  $L_{\text{CM}} = \text{cons.}$

even if CM is moving  
or accelerating

Ex 3.4



$$\dot{\vec{P}} = \vec{F}_{\text{ext}}, \quad \vec{P} = \vec{F} \Delta t$$

$$\vec{P} = M \dot{\vec{R}} = (2m) \dot{\vec{R}} = \vec{F} \Delta t$$

$$\vec{V}_{\text{cm}} = \dot{\vec{R}} = \vec{F} \Delta t / 2m$$

$$\vec{P}_{\text{ext}} = \vec{F} b$$

$$\frac{d}{dt} L = \overline{F}_{ext}$$

$$L = \overline{F}_{ext} dt$$

$$L = Fb \Delta t$$

$$L = I\omega, \quad I = 2mb^2$$

$$L = 2mb^2 \omega$$

$$\omega = F \Delta t / 2mb$$

$$V_{left} = V_{cm} + \omega b = \frac{F \Delta t}{m}$$

$$V_{right} = V_{cm} - \omega b = 0$$